Training Course

Module 08: PaaS Compute Options

Agenda

Lesson 01: Configure Azure App Service Plans

Lesson 02: Configure Azure App Services

Lesson 03: Configure Azure Container Instances

Lesson 04: Configure Azure Kubernetes Service

Lesson 01: Configure Azure App Service Plans

Implement Azure App Service Plans

Define a set of compute resources for a web app to run

Determines performance, price, and features

One or more apps can be configured to run in the same App Service plan

Region where compute resources will be created

Number of virtual machine instances

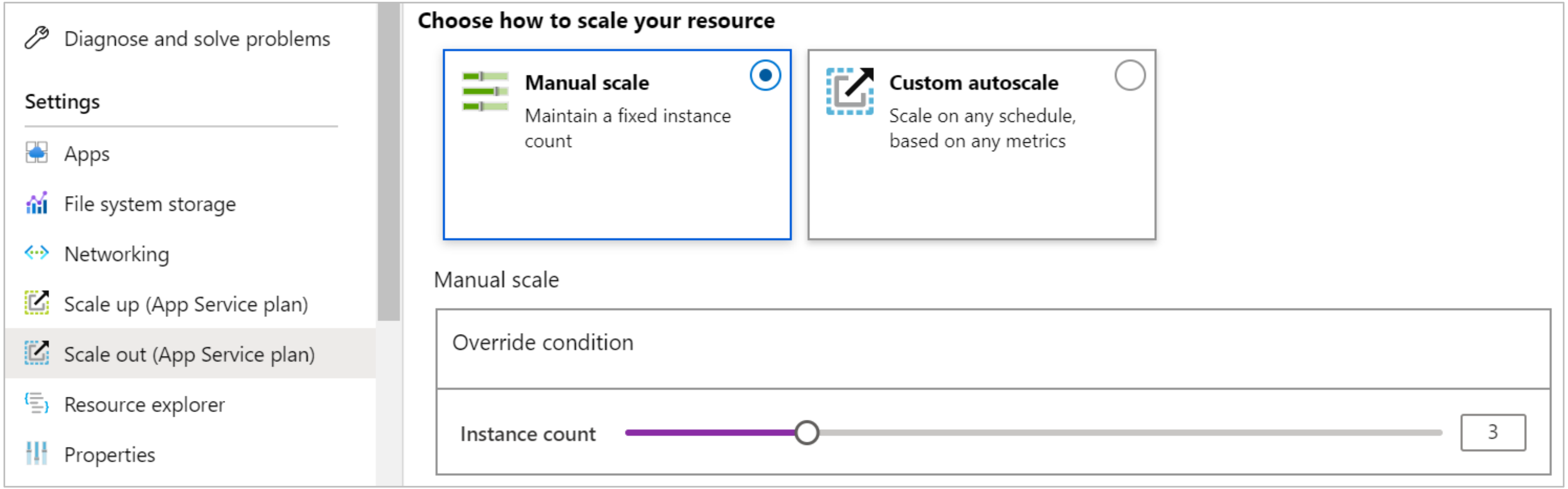
Size of virtual machine instances

Pricing tier (next slide)

Determine App Service Plan Pricing

Selected Features	Free	Shared (dev/test)	Basic (dedicated dev/test)	Standard (production workloads)	Premium (enhanced scale and performance)	Isolated (high-performance, security and isolation)
Web, mobile, or API apps	10	100	Unlimited	Unlimited	Unlimited	Unlimited
Disk space	1 GB	1 GB	10 GB	50 GB	250 GB	1 TB
Auto Scale	–	–	–	Supported	Supported	Supported
Deployment Slots	0	0	0	5	20	20
Max Instances	–	–	Up to 3	Up to 10	Up to 30	Up to 100

Scale Up and Scale Out the App Service Plan



Scale up (change the App Service plan):

More hardware (CPU, memory, disk)

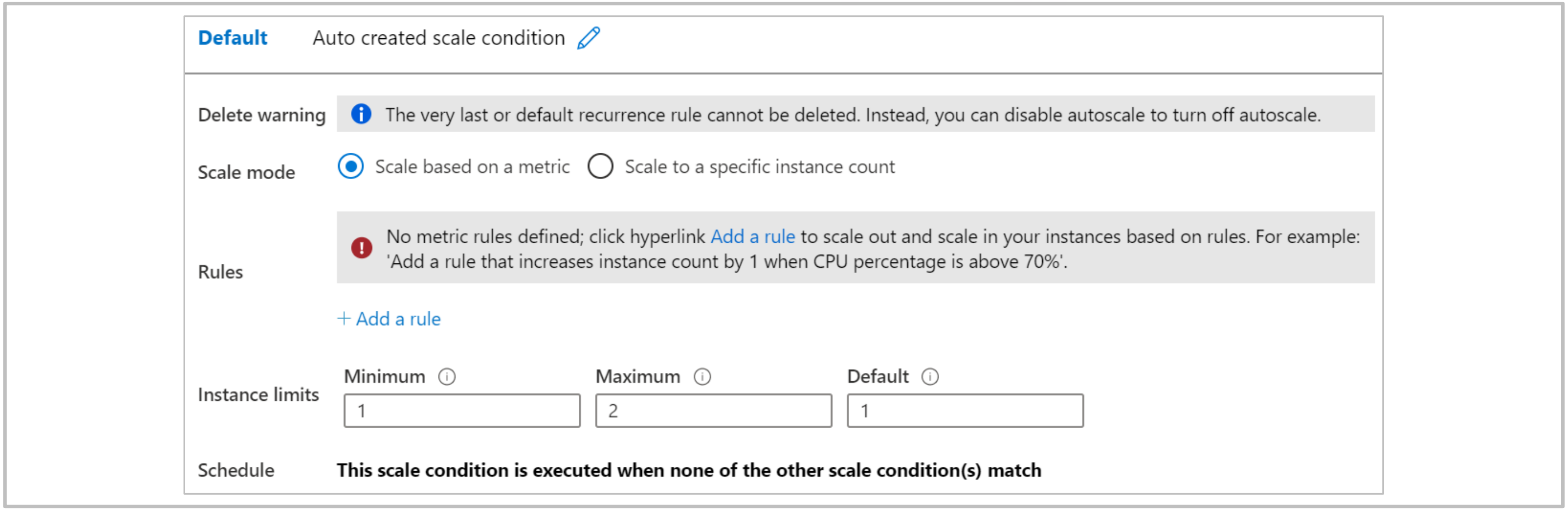
More features (dedicated virtual machines, staging slots, autoscaling)

Scale out (increase the number of VM instances):

Manual (fixed number of instances)

Auto scale (based on predefined rules and schedules)

Configure App Service Plan Scaling



Adjust available resources based on the current demand

Improves availability and fault tolerance

Scale based on a metric (CPU percentage, memory percentage, HTTP requests)

Scale according to a schedule (weekdays, weekends, times, holidays)

Can implement multiple rules – combine metrics and schedules

Don't forget to scale in

Lesson 02: Configure Azure App Services

Implement Azure App Service

Fpt Software | CAMPUS

.NET | Node.js | PHP | Java | Python (on Linux) | HTML | Custom Windows/Linux Container

Includes Web Apps, API Apps, Mobile Apps, and Function Apps

Fully managed environment enabling high productivity development

Platform-as-a-service (PaaS) offering for building and deploying highly available cloud apps for web and mobile

Platform handles infrastructure so developers focus on core web apps and services

Developer productivity using .NET, .NET Core, Java, Python and a host of others

Provides enterprise-grade security and compliance

Create an App Service

Name must be unique

Access using *azurewebsites.net* – can map to a custom domain

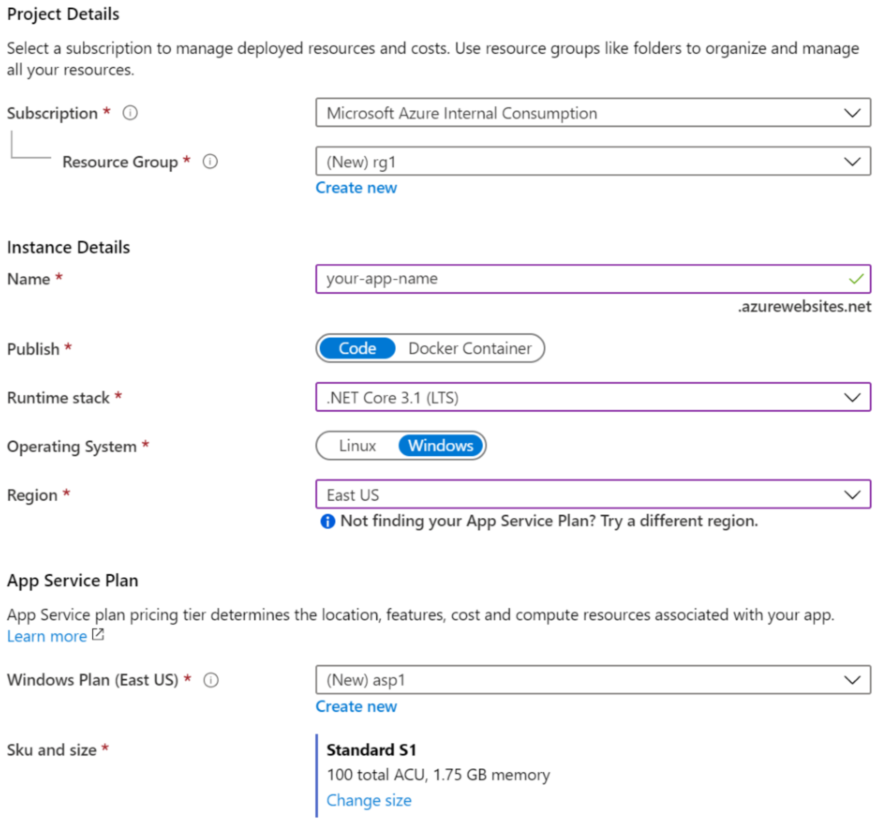
Publish Code (Runtime Stack)

Publish Docker Container

Linux or Windows

Region closest to your users

App Service Plan



Continuous Integration and Deployment

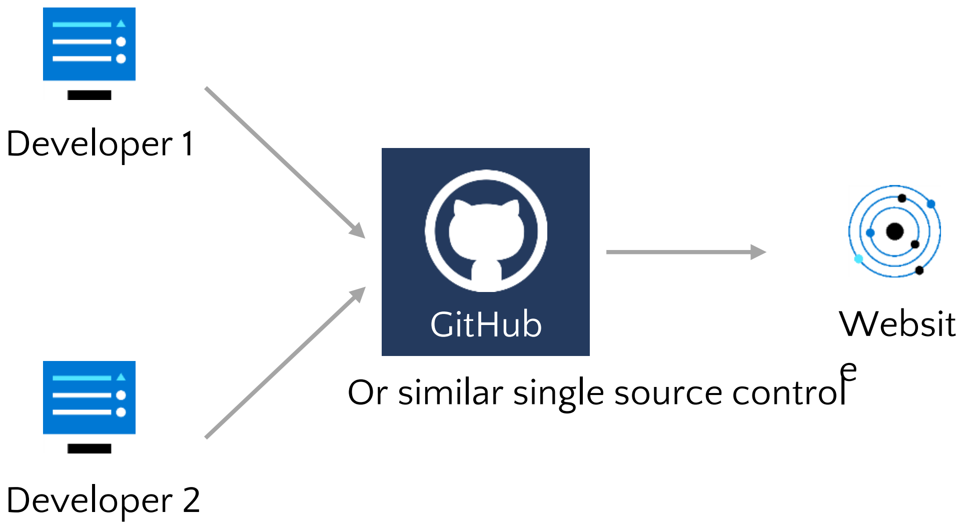
Work in a single source control

Whenever code updates are pushed to the source control, then the website or web app will automatically pick up the updates

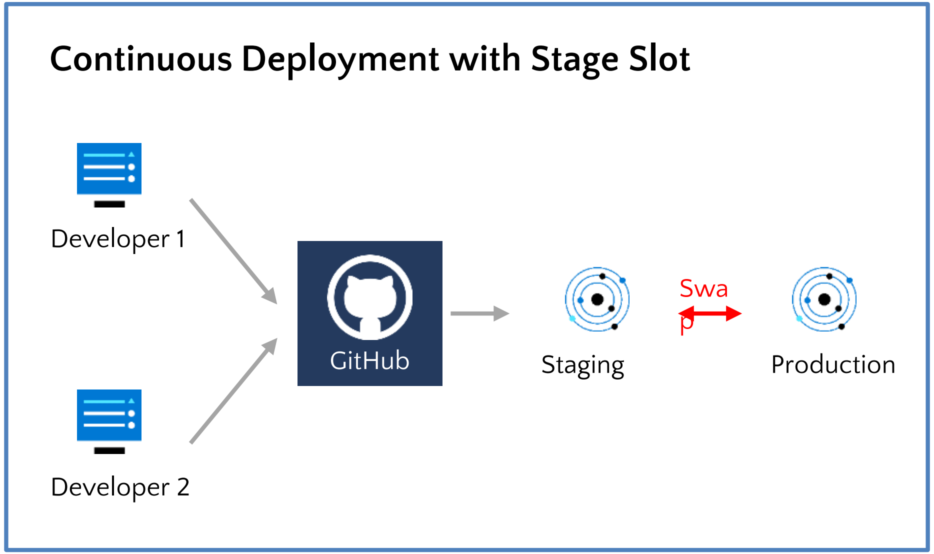
A continuous deployment workflow publishes the most recent updates from a project

Use the portal for continuous deployments from GitHub, Bitbucket, or Azure DevOps

Continuous Deployment



Create Deployment Slots



Service Plan	Slots
Free, Shared, Basic	0
Standard	Up to 5
Premium	Up to 20
Isolated	Up to 20

- Deploy to a different deployment slots (depends on service plan)
- Validate changes before sending to production
- Deployment slots are live apps with their own hostnames
- Avoids a cold start – eliminates downtime
- Fallback to a last known good site
- Auto Swap when pre-swap validation is not needed

Add Deployment Slots

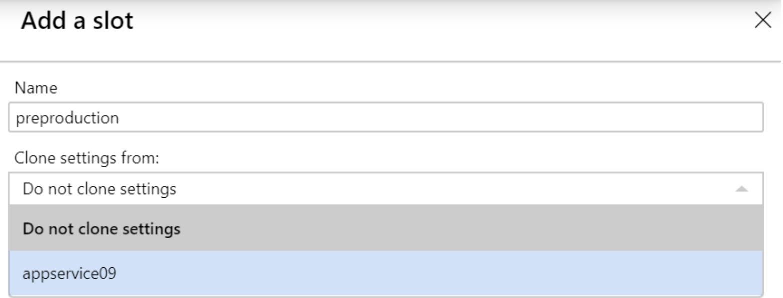
Select whether to clone an app configuration from another deployment slot

When you clone, pay attention to the settings:

- Slot-specific app settings and connection strings
- Continuous deployment settings
- App Service authentication settings

Not all settings are sticky (endpoints, custom domain names, SSL certificates, scaling)

Review and edit your settings before swapping



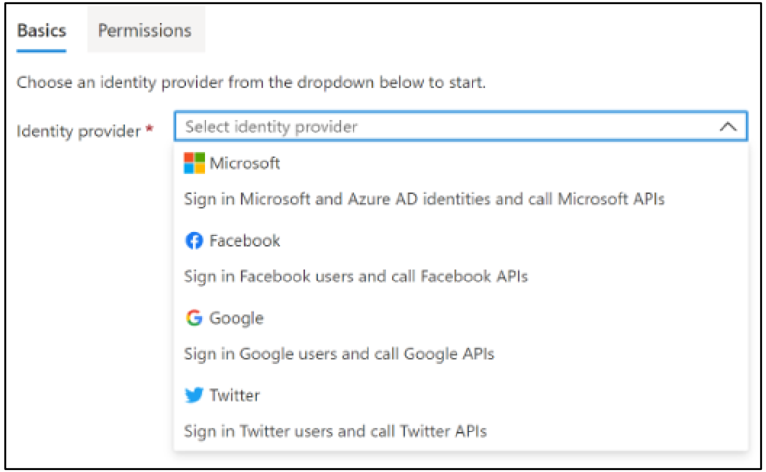
Secure an App Service

Authentication:

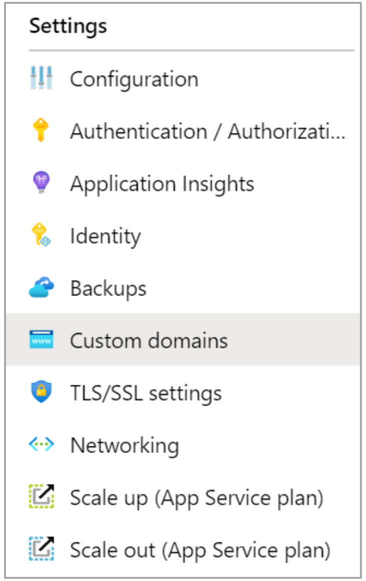
- Enable authentication – default anonymous
- Log in with a third-party identity provider

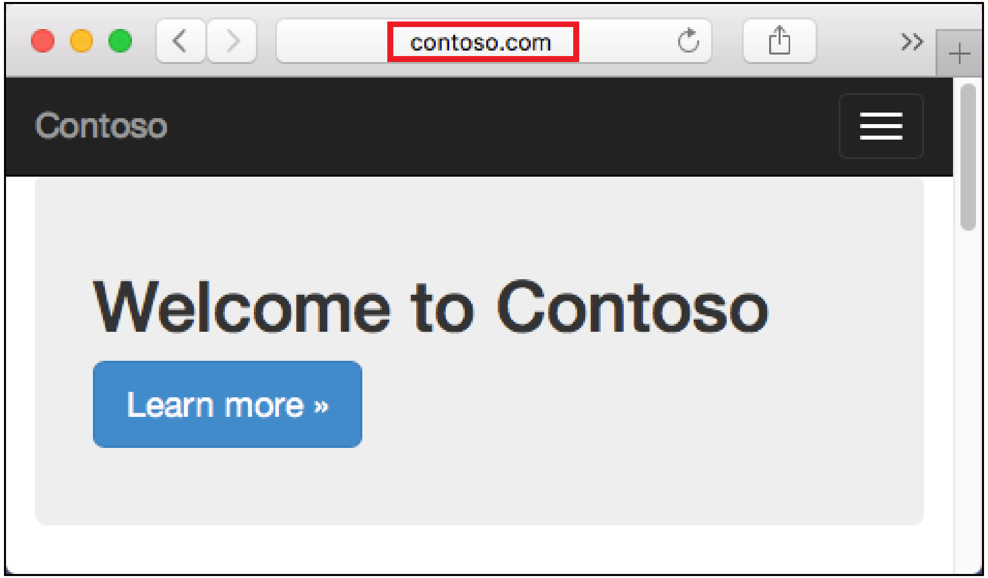
Security:

- Troubleshoot with Diagnostic Logs – failed requests, app logging
- Add an SSL certificate – HTTPS
- Define a priority ordered allow/deny list to control network access to the app
- Store secrets in the Azure Key Vault



Create Custom Domain Names

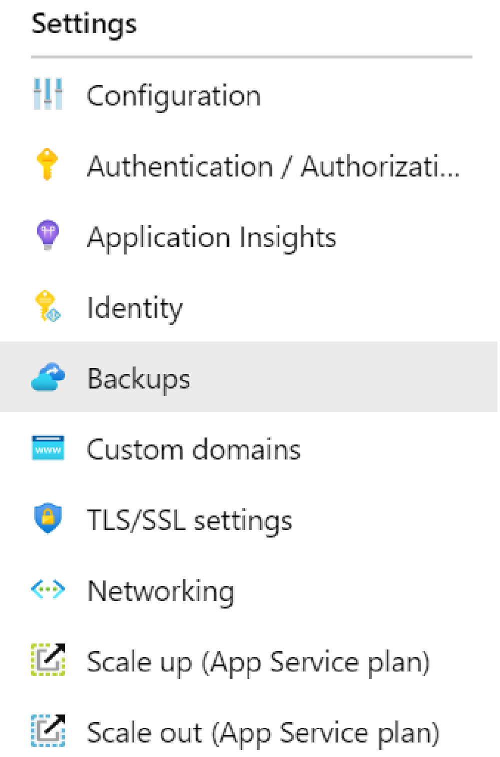




Redirect the default web app URL	Validate the custom domain in Azure	Use the DNS registry for your domain provider – create a CNAME or A record with the mapping	Ensure App Service plan supports custom domains

Backup an App Service

- Create app backups manually or on a schedule
- Backup the configuration, file content, and database connected to the app
- Requires Standard or Premium plan
- Backups can be up to 10 GB of app and database content
- Configure partial backups and exclude items from the backup
- Restore your app on-demand to a previous state, or create a new app



Use Application Insights

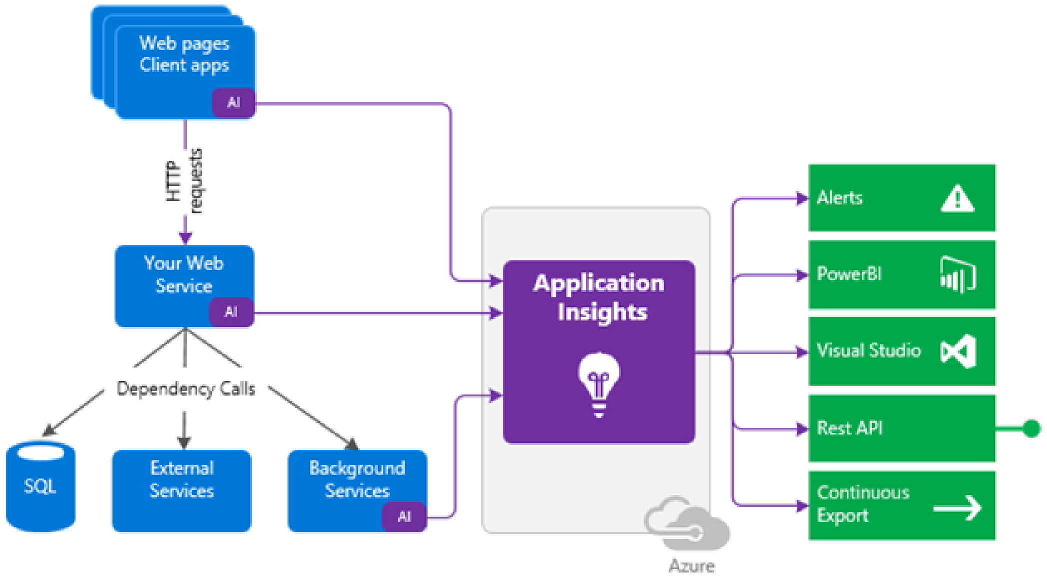
Request rates, deny rates, response time and failure rates

Page views and load performance

User and session counts

Performance counters

Diagnostics and Exceptions



Lesson 03: Configure Azure Container Instances

Compare Containers to Virtual Machines

Feature	Containers	Virtual Machines
Isolation	Typically provides lightweight isolation from the host and other containers but doesn't provide as strong a security boundary as a virtual machine	Provides complete isolation from the host operating system and other VMs. This is useful when a strong security boundary is critical, such as hosting apps from competing companies on the same server or cluster
Operating system	Runs the user mode portion of an operating system and can be tailored to contain just the needed services for your app, using fewer system resources.	Runs a complete operating system including the kernel, thus requiring more system resources (CPU, memory, and storage)
Deployment	Deploy individual containers by using Docker via command line; deploy multiple containers by using an orchestrator such as Azure Kubernetes Service	Deploy individual VMs by using Windows Admin Center or Hyper-V Manager; deploy multiple VMs by using PowerShell or System Center Virtual Machine Manager
Persistent storage	Use Azure Disks for local storage for a single node, or Azure Files (SMB shares) for storage shared by multiple nodes or servers	Use a virtual hard disk (VHD) for local storage for a single VM, or an SMB file share for storage shared by multiple servers
Fault tolerance	If a cluster node fails, any containers running on it are rapidly recreated by the orchestrator on another cluster node	VMs can fail over to another server in a cluster, with the VM's operating system restarting on the new server

Explore Azure Container Instances Benefits

PaaS Service

Fast startup times

Public IP connectivity and DNS name

Isolation features

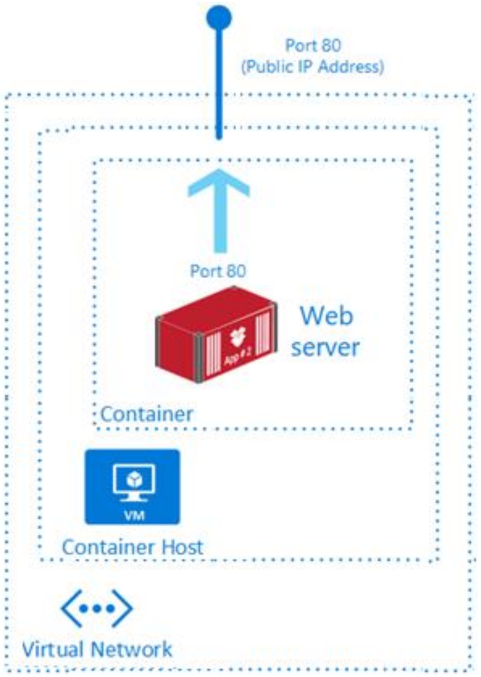
Custom sizes

Persistent storage

Linux and Windows Containers

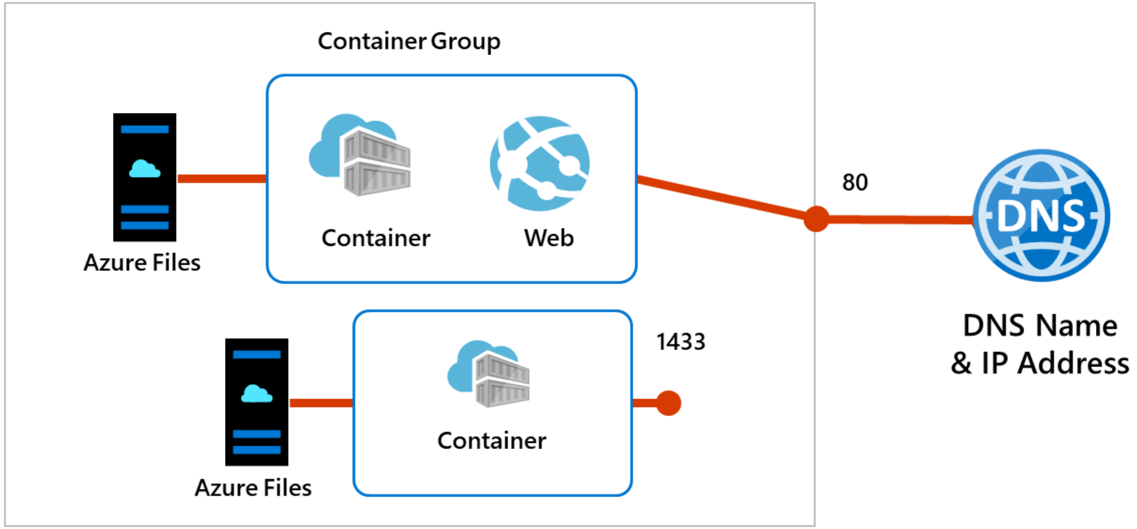
Co-scheduled Groups

Virtual network Deployment



Fastest way to run a container in Azure without provisioning a VM

Implement Container Groups

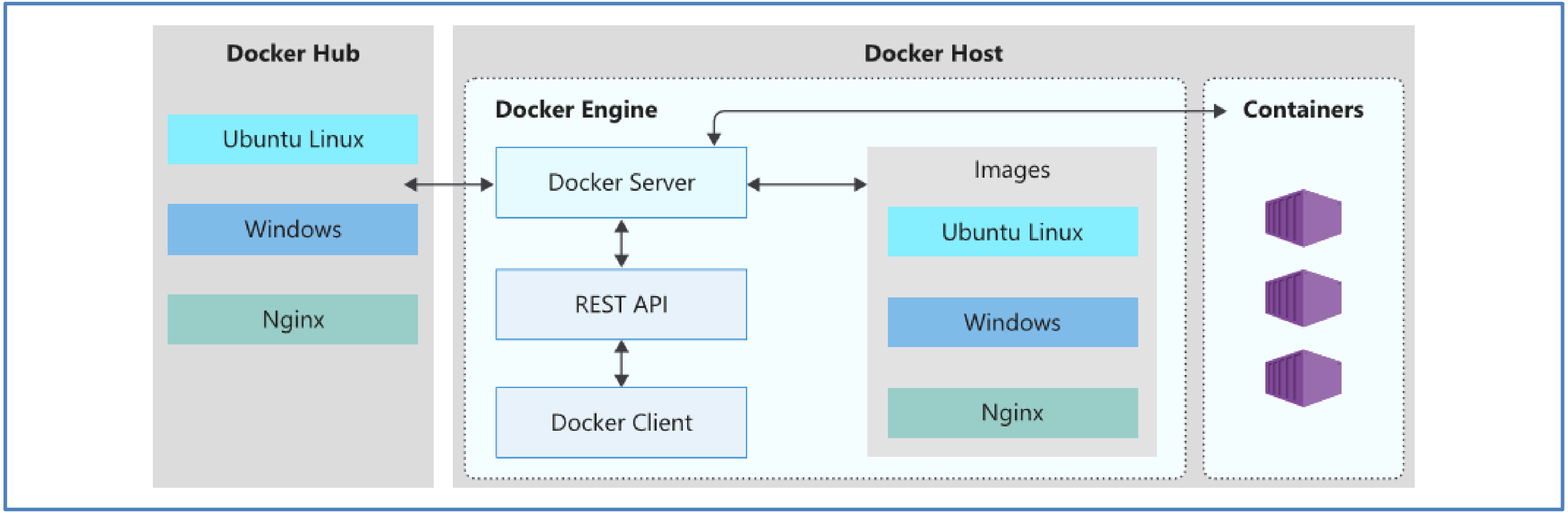


Top-level resource in Azure Container Instances

A collection of containers that get scheduled on the same host

The containers in the group share a lifecycle, resources, local network, and storage volumes

Understand the Docker Platform



Enables developers to host applications within a container

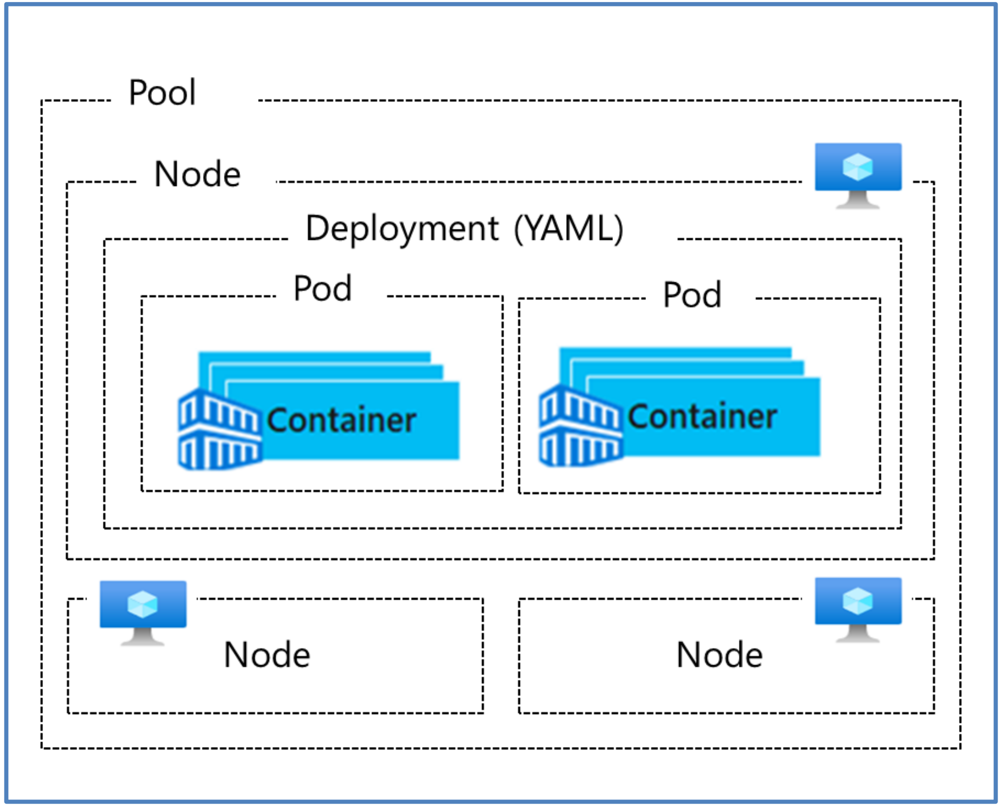
A container is a standardized "unit of software" that contains everything required for an application to run

Available on both Linux and Windows and can be hosted on Azure

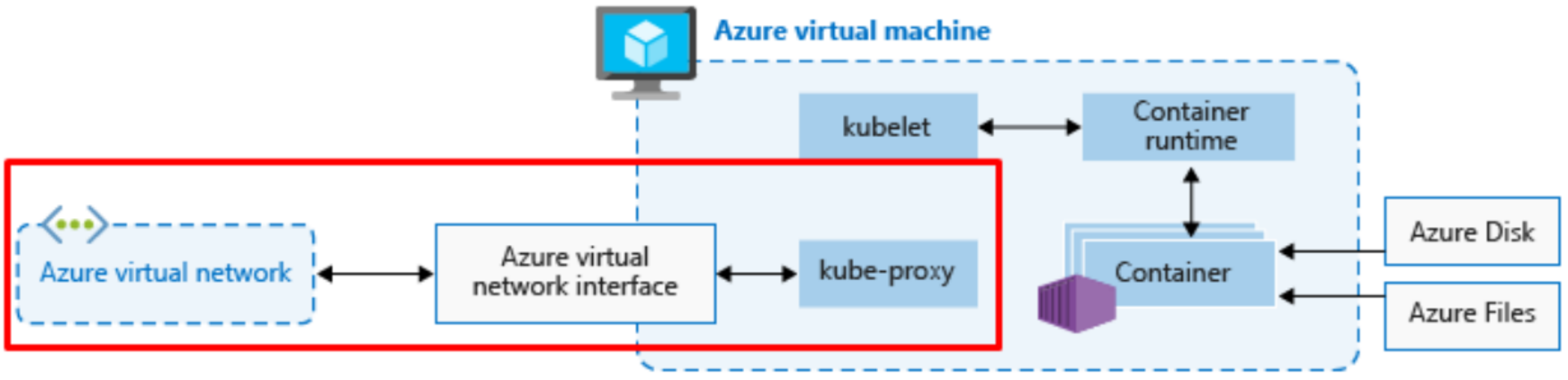
Lesson 04: Configure Azure Kubernetes Service

Understand AKS Terminology

Term	Description
Pools	Groups of nodes with identical configurations
Nodes	Individual VMs running containerized applications
Pods	Single instance of an application. A pod can contain multiple containers
Deployment	One or more identical pods managed by Kubernetes
Manifest	YAML file describing a deployment



Understand AKS Clusters and Nodes

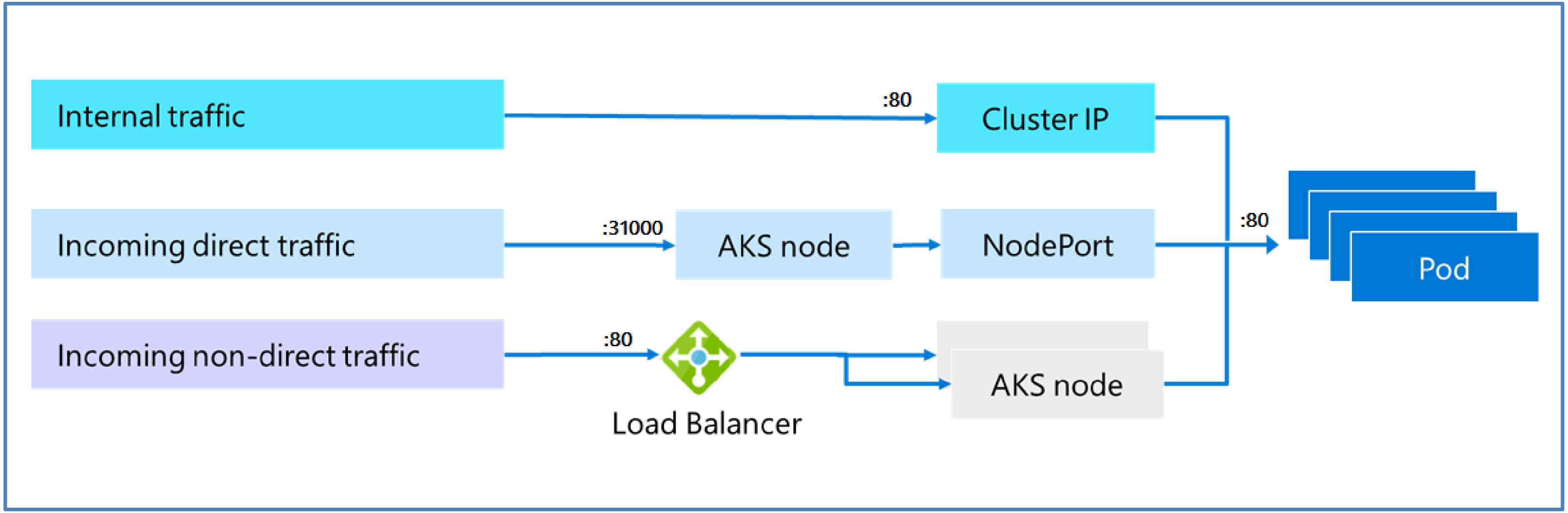


Azure-managed node provides core Kubernetes services and orchestration

Customer-managed nodes run applications and supporting services

Each individual node is an Azure virtual machine

Configure AKS Networking



Pods run an instance of your application

Services group pods together to provide network connectivity

ClusterIP provides internal traffic access

NodePort provides mapping for incoming direct traffic

LoadBalancer has external IP address for incoming non-direct traffic

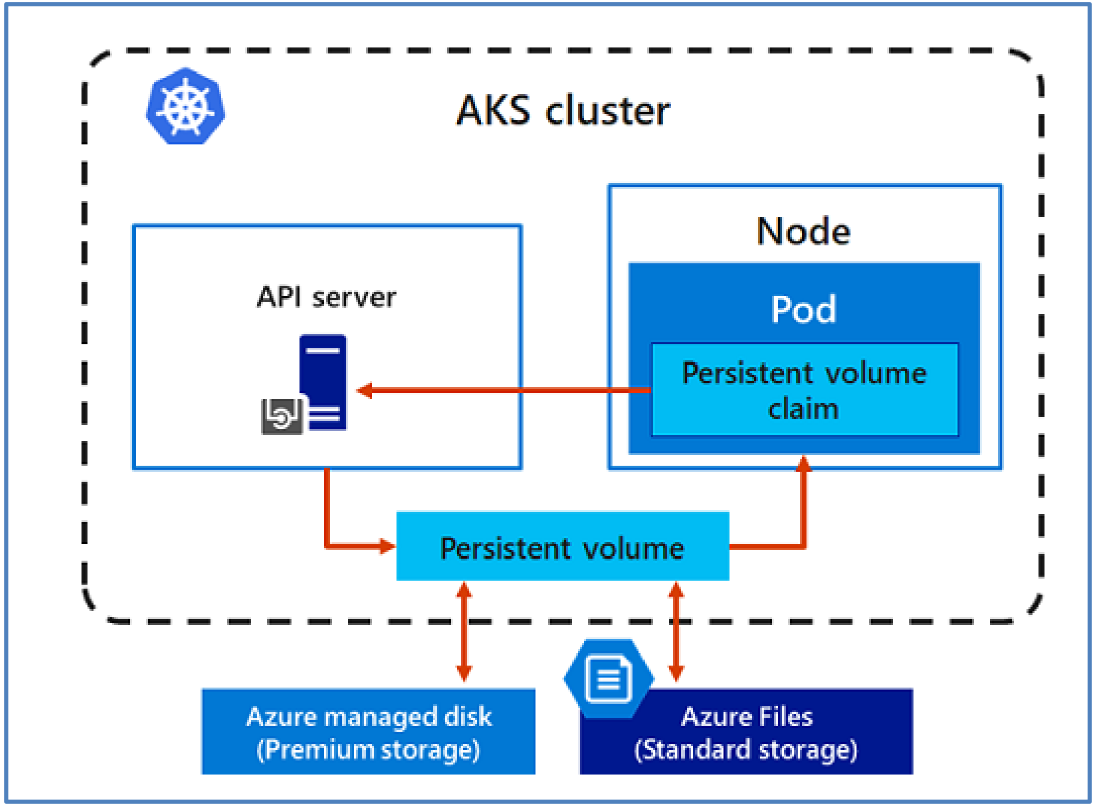
Configure AKS Storage

Local storage on the node is fast and simple to use

Local storage might not be available after the pod is deleted

Multiple pods may share data volumes

Storage could potentially be reattached to another pod



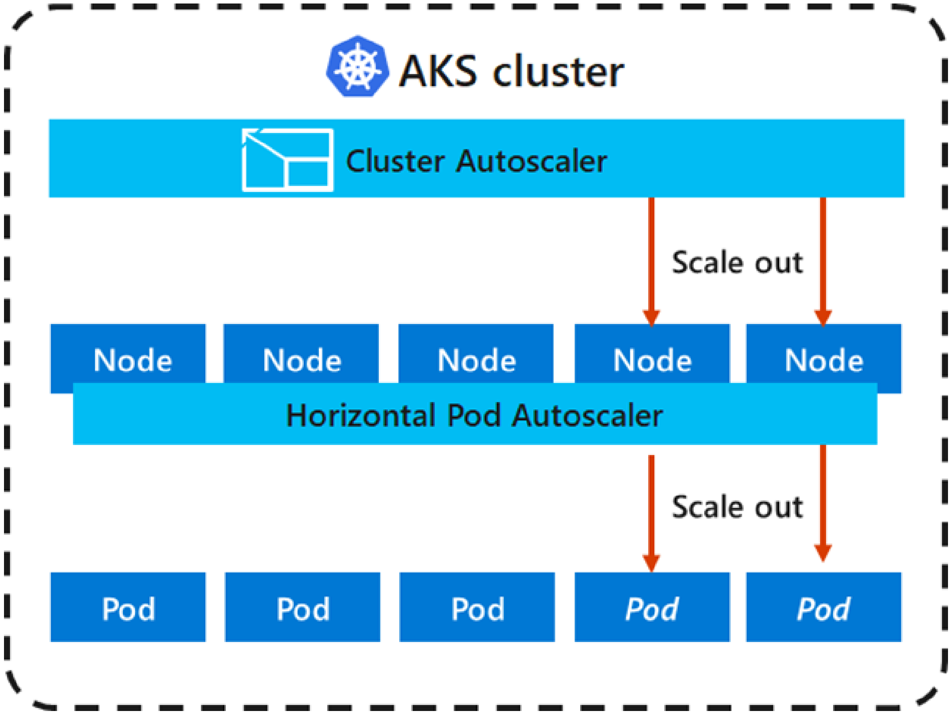
Configure AKS Scaling

Applications might grow beyond the capacity of a single pod

Kubernetes has built-in autoscalers

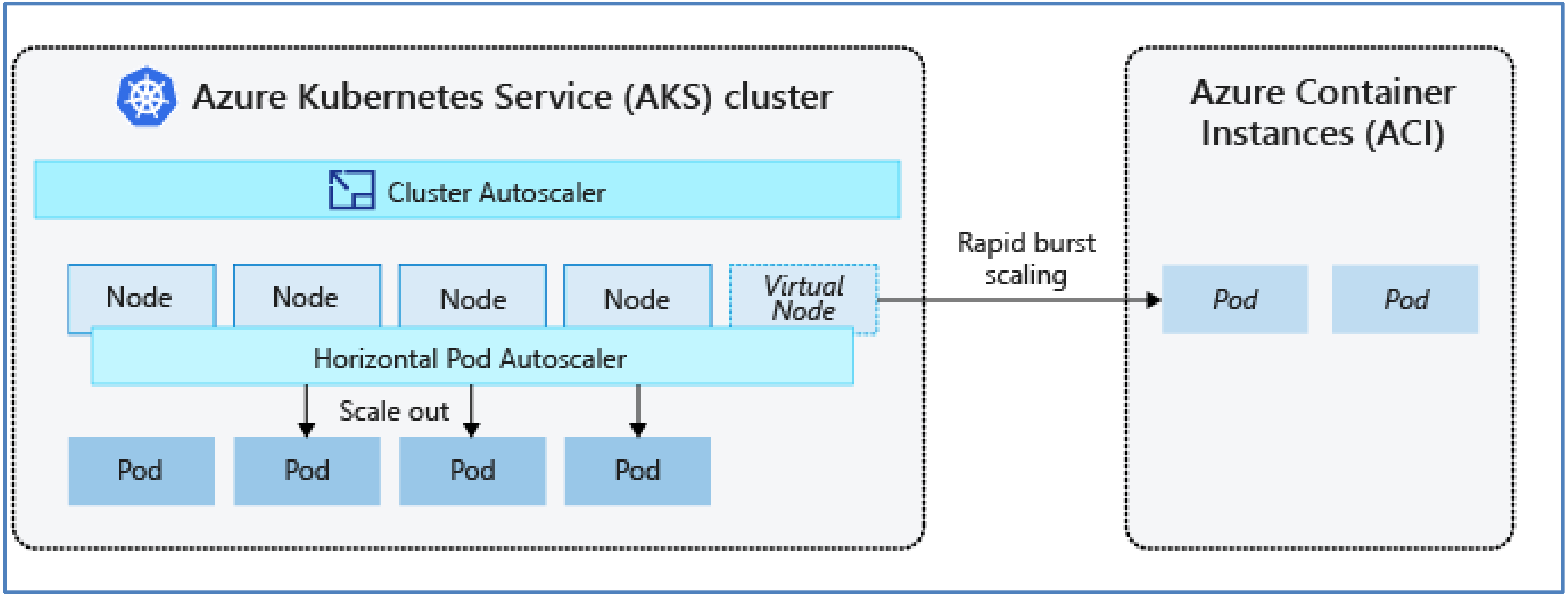
Cluster autoscaler scales based on compute resources

Horizontal pod autoscaler scales based on metrics

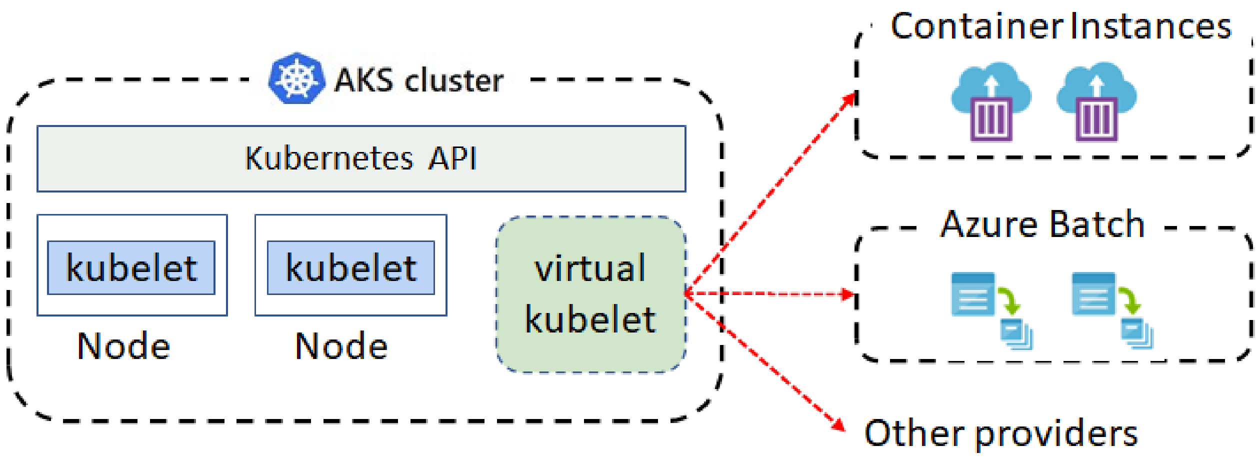


Configure AKS Scaling to ACI

If you need to rapidly grow your AKS cluster, you can create new pods in Azure Container Instances



Virtual Kubelet

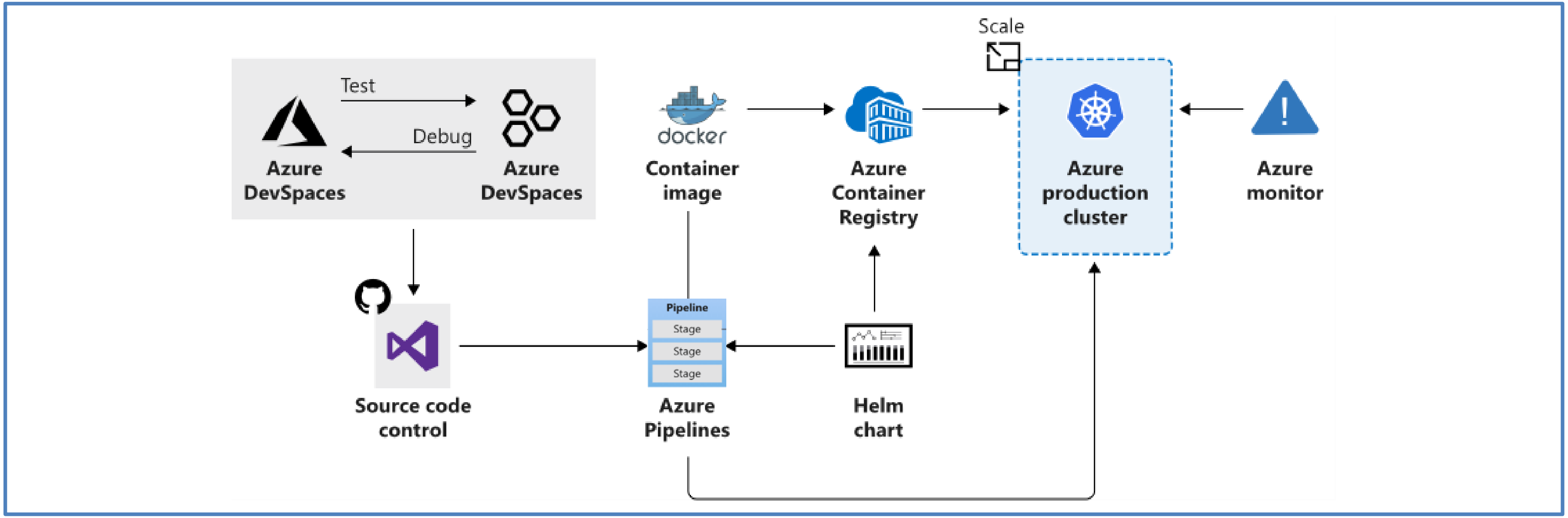


Virtual kubelet is an open-source Kubernetes kubelet implementation

The virtual kubelet registers itself as a node and allows developers to deploy pods and containers with their own APIs

Supported by an ecosystem of providers

Azure Kubernetes Service



Manages health monitoring and maintenance

Performs simple cluster scaling

Enables nodes to be fully managed by Microsoft

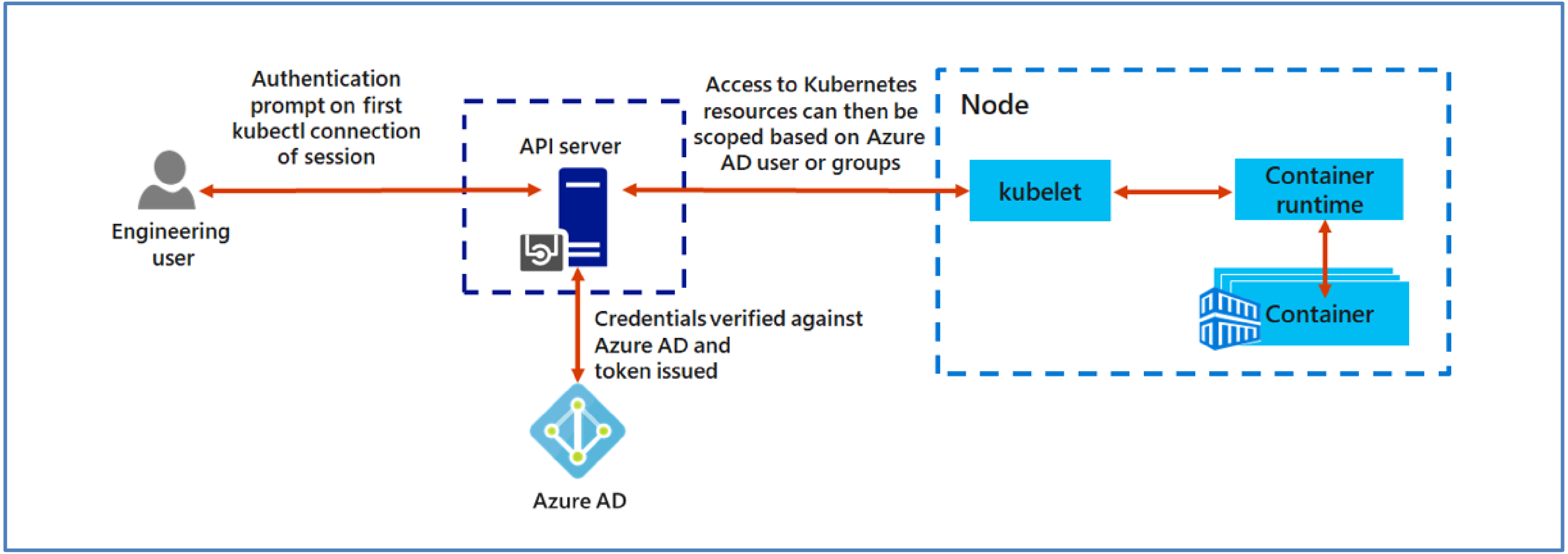
You're responsible only for managing the agent nodes

You pay only for the agent nodes

Connect AKS and Azure Active Directory

Use Azure AD as an integrated identity solution

Use service accounts, user accounts, and role-based access control



Configure AKS Security

Implement security across the entire AKS infrastructure

Managed service – Limit access with authorized IP ranges, create a private cluster, use RBAC and Azure AD access

Cluster upgrades – Upgrade the AKS cluster with cordon and drain

Node – Automatic OS security patches, Azure managed disks, pod security policies

Networks – Define ingress controllers with private internal IP addresses, filter the flow of traffic with network security groups

Data – Kubernetes secrets for credentials and keys

Thank you